FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Advanced)-2020

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **One Section**: **Section-A.**

1. ***Section–A (01 – 07, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (08– 14, 26 – 32, 44 – 50) *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*
   Partial Marks **+1** for each correct option provided no incorrect option is selected.

   Section–A (15 – 18, 33 – 36, 51 – 54) *contains 6 paragraphs. Each paragraph is having 2 multiple choice questions having 4 options with* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. In the Young's double slit experiment using a monochromatic light of wavelength λ, the path difference (in terms of an integer n) corresponding to any point having half the peak intensity is

(A) $(2n+1)\frac{\lambda}{2}$ (B) $(2n+1)\frac{\lambda}{4}$

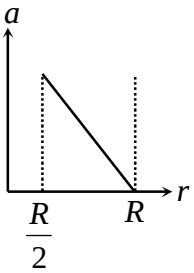
(C) $(2n+1)\frac{\lambda}{8}$ (D) $(2n+1)\frac{\lambda}{16}$

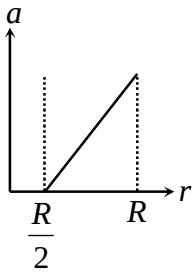
2. The speed of a wave on a string is 150 m/s when the tension is 120 N. The percentage increase in the tension in order to raise the wave speed by 20% is:

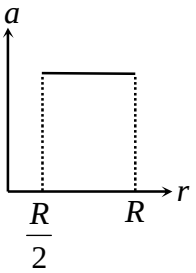
(A) 44% (B) 40%

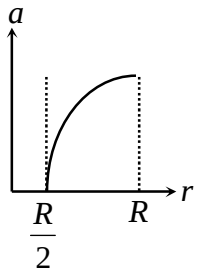
(C) 20% (D) 10%

3. A frictionless tunnel is dug along a chord of the earth at a perpendicular distance $\frac{R}{2}$ from the centre of earth (where R is radius of earth). An object is released from one end of the tunnel. The correct graph, showing the variation of acceleration of particle with its distance r from centre of earth is

(A)


(B)


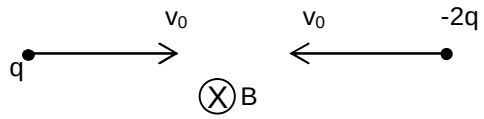
(C)


(D)


Space for Rough work

4. A closed organ pipe of length *L* is vibrating in its first overtone. There is a point *Q* inside the pipe at a distance 7*L*/9 from the open end. The ratio of pressure amplitude at *Q* to the maximum pressure amplitude in the pipe is
(A) 1 : 2 (B) 2 : 1
(C) 1 : 1 (D) 2 : 3

5. A charge particle q of mass m is placed at a distance d from another charge particle – 2q of mass 2m in a uniform magnetic field B as shown. If particles are projected towards each other with equal speed v_0, so that the two particles touches each other without collision during its motion. (Assume only force due to magnetic field acts on the particle)



(A) $\frac{qBd}{m}$ (B) $\frac{qBd}{2m}$
(C) $\frac{2qBd}{m}$ (D) $\frac{3qBd}{2m}$

6. When the electron in a hydrogen atom jumps from the second orbit to the first orbit, the wavelength of the radiation emitted is λ. When the electron jumps from the third to the first orbit, the wavelength of the radiation emitted as
(A) $\frac{9}{4}\lambda$ (B) $\frac{4}{9}\lambda$
(C) $\frac{27}{32}\lambda$ (D) $\frac{32}{27}\lambda$

7. The image of an object, formed by a plano-convex lens at a distance of 8 m behind the lens, is real and is one-third the size of the object. The wavelength of light inside the lens is $\frac{2}{3}$ times the wavelength in free space. The radius of the curved surface of the lens is
(A) 1 m (B) 2 m
(C) 3 m (D) 6 m

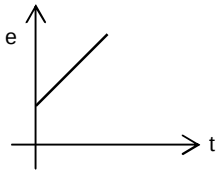
Space for Rough work

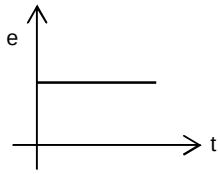
(One or More than one correct type)

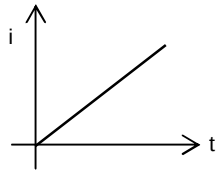
This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

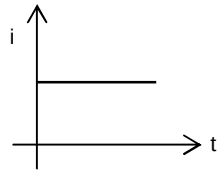
8. A bimetallic strip is formed by two identical strips, one of copper and the other of brass. The coefficients of linear expansion of the two metals are α_C and α_B. On heating, the temperature of the strip goes up by ΔT and the strip bends to from an arc of radius of curvature R. Then R is
(A) proportional to ΔT (B) inversely proportional to ΔT
(C) proportional to $|\alpha_B - \alpha_C|$ (D) inversely proportional to $|\alpha_B - \alpha_C|$

9. Two parallel long straight conductors lie on a smooth horizontal surface. Two other parallel conductors rest on them at right angles so as to form a square of side a initially. A uniform magnetic field B exists in vertical direction. Now all the four conductors start moving outwards with a constant velocity v. The induced e.m.f. e and induced current i will vary with time t as

(A) 

(B) 

(C) 

(D) 

Space for Rough work

10. Consider the decay $^{238}Pu \rightarrow ^{234}U + ^{4}He$
The atomic masses are given as
$^{238}_{238.04955u}Pu$ $^{234}_{234.04095u}U$ $^{4}_{4.002633u}He$
Neglecting the recoil of the residual nucleus
(A) the kinetic energy of the α particle emitted is 5.58 MeV
(B) the kinetic energy of the α particle emitted is 5.58 eV
(C) conservation of electric charge is valid for the given reaction
(D) conservation of electric charge is not valid for the given reaction

11. In a thermodynamic process Helium gas obeys the law $TP^{-2/5}$ constant. If temperature of 2 moles of the gas is raised from T to 3T, then
(A) Heat given to the gas is 9 RT
(B) Heat given t the gas zero
(C) Work done by the gas is $-6RT$
(D) $\int VdP$ for the process in the given temperature range is 10RT

12. Sounds from two identical sources S_1 and S_2 reach a point P. when the sounds reach directly and in the same phase the intensity at P is I_0. The power of S_1 is now reduced by 64% and the phase difference between S_1 and S_2 is varied continuously the maximum and minimum intensities recorded at P are now I_{max} and I_{min}
(A) $I_{max} = 0.64\ I_0$ (B) $I_{min} = 0.36\ I_0$
(C) $I_{max}/I_{min} = 16$ (D) $I_{max}/I_{min} = 1.64/0.36$

13. A wave disturbance in a medium is described by
$y(x, t) = 0.02 \cos (50\ \pi t + \pi/2) \cos (10\ \pi x)$,
where 'x' and 'y' are in metre and 't' in seconds.
(A) A node occurs at x = 0.15 m
(B) An antinode occurs at x = 0.3 m
(C) The speed of the component wave is 5.0 m/s
(D) The wavelength is 0.2 m.

Space for Rough work

14. A cannon shell is fired to hit a target at a horizontal distance R, however it breaks into two equal parts at its highest point, one part returns to the cannon. The other part
(A) will fall at a distance R beyond target (B) will fall at a distance 3R beyond target
(C) will hit the target (D) have nine times kinetic energy of first

Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 15 and 16

In a photoelectric effect, a point source of green light of power 40 W emits mono-energetic photons that can just emit photoelectrons from an isolated metallic sphere of radius r = 1 cm, placed at a distance of 1m from the light source. Now, four other separate sources of violet, blue, red and yellow light emitting same number of photons as that of green light are brought near the source of green light.

Assume one photoelectron is emitted out of every 10^6 incident photons on sphere.
(Take λ_{violet} = 4136 Å, λ_{blue} = 5000 Å, λ_{red} = 7200 Å, λ_{green} = 4963 Å, hc = 12408 eV Å)

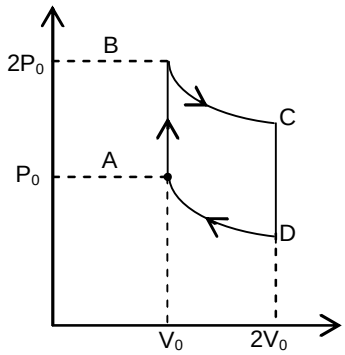
15. No. of photoelectrons emitted from the sphere per second is
(A) 2.5×10^9 (B) 2.5×10^8
(C) 2.5×10^7 (D) 2.5×10^6

16. The potential of the sphere when emission of photoelectrons will stop is
(A) 0.7 V (B) 0.5 V
(C) 0.6 V (D) 0.8 V

Space for Rough work

Paragraph for Question Nos. 17 and 18

One mole of a monatomic gas undergoes the cycle shown in the diagram: The pressure, volume and temperature at A are P_0, V_0 and T_0 respectively; the pressure at B is $2P_0$ and the volume at D is $2V_0$. The processes AB and CD are isochoric. During BC DA, the gas undergoes the processes such that during an infinitesimal contraction or expansion, the heat absorbed dQ, the change in internal energy dU and dW, the work done satisfy the relation:



$$dQ = \frac{1}{2}dU + \frac{2}{3}dW$$

For the process BC, DA, we can write

$$dQ = \frac{1}{2}dU + \frac{2}{3}dW = dU + dW$$

$$\text{or, } +\frac{1}{2}dU = -\frac{1}{3}dW \quad ; \quad \text{or, } \frac{1}{2}\cdot\frac{3}{2}RdT = -\frac{1}{3}dW$$

$$pdV + Vdp \Rightarrow RdT \quad ; \quad \text{or, } \frac{13}{9}pdV + Vdp = 0$$

$$\text{or, } pV^{\frac{13}{9}} = \text{constant}$$

17. Find the temperature at C
(A) $2T_0$ (B) $3T_0$
(C) $2^{5/9}T_0$ (D) $3^{5/9}T_0$

18. What is the efficiency of the cycle? Assume $2^{5/9} = 1.5, 2^{4/9} = 4/3$ approximately
(A) 24% (B) 30%
(C) 36% (D) 42%

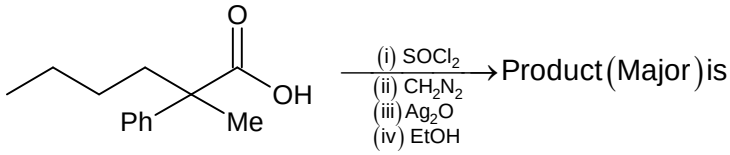
Space for Rough work

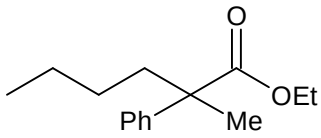
Chemistry

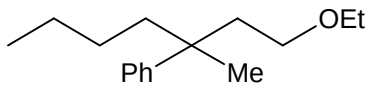
PART – II

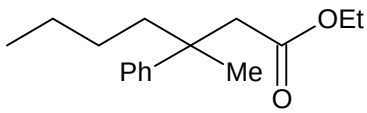
SECTION – A
(One Options Correct Type)

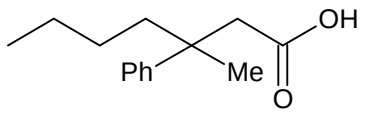
This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. 

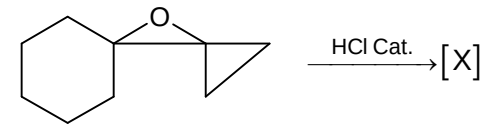
(A) 

(B) 

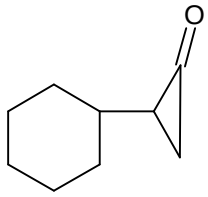
(C) 

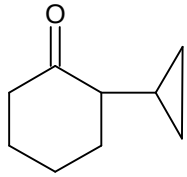
(D) 

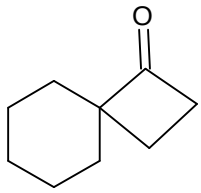
20. Which one of the following reagent can be used for differentiating the Cu^{2+} and Bi^{3+}?
(A) H_2S gas in presence of dil HCl (B) NH_4OH(excess)
(C) $K_4[Fe(CN)_6]$ (D) (B) and (C) both

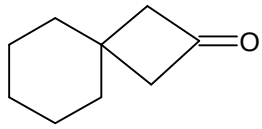
21. 

The compound[X] is

(A) 

(B) 

(C) 

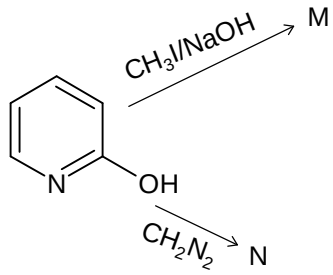
(D) 

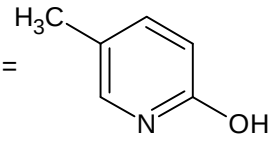
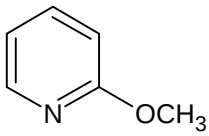
Space for Rough work

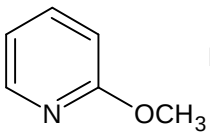
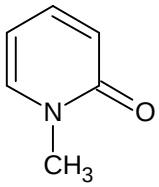
22. The compound formed by dissolving gold and platinum in aqua-regia is
(A) $[AuCl_4]^{3-}$ and $[PtCl_6]^{2-}$ (B) $[AuCl_4]^{-}$ and $[PtCl_6]^{4-}$
(C) $[AuCl_4]^{-}$ and $[PtCl_6]^{2-}$ (D) $[AuCl_4]^{-}$ and $[PtCl_6]^{2-}$

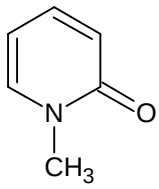
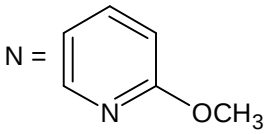
23. The pair of compounds having the same hybridization for the central atom is
(A) $[Cu(NH_3)_4]^{2+}$ and $[Ni(NH_3)_4]^{2+}$ (B) $[NiCl_4]^{2-}$ and $[PtCl_4]^{2-}$
(C) $[Cu(NH_3)_4]^{2+}$ and $[Zn(NH_3)_4]^{2+}$ (D) $[Co(NH_3)_6]^{3+}$ and $[Co(H_2O)_6]^{3+}$

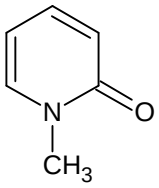
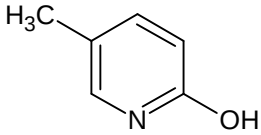
24. The major products M and N formed in the following reactions are:–



(A) M =  N = 

(B) M =  N = 

(C) M =  N = 

(D) M =  N = 

Space for Rough work

25. The highest occupied MO in N_2 and O_2^+ respectively are (take x-axis as inter nuclear axis)
(A) σ2px, π^*2py (B) π2py, π2pz
(C) σ^*2px, σ2px (D) π^*2py, π^*2pz

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

26. Which of the following statement(s) are correct?
(A) $CH_3 - Cl > CH_3 - F > CH_3 - Br > CH_3 - I$ (Order of dipole moment)
(B) $Cl_2 > F_2 > Br_2 > I_2$ (Order of bond energy)
(C) $S_2O_3^{2-}, S_2O_4^{2-}, S_2O_5^{2-}$ (All have S – S linkage)
(D) $PCl_5 > SiCl_4 > AlCl_3 > MgCl_2$ (Order of hydrolysis)

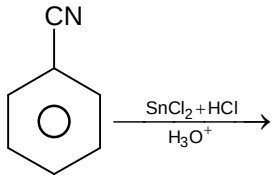
27. Which of the following statements is/are true about the transition metal alkene complexes?
K [Pt $Cl_3C_2H_4$]
(A) Back bonding weakens the double bond of the alkene
(B) σ bonding and back-bonding synergistically strengthen metal alkene interaction
(C) Electron withdrawing substituents on alkene reduce back bonding
(D) π–acidic co-ligands on metal strengthen back bonding

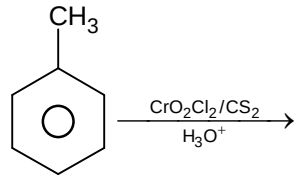
28. On electrolysis, in which of the following, O_2 would be liberated at the anode?
(A) dil H_2SO_4 with Pt electrode (B) aq. $AgNO_3$ with Pt electrode
(C) dil H_2SO_4 with Cu electrode (D) aq. NaOH with Pt electrode

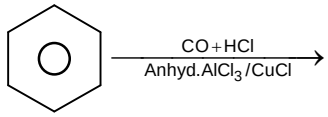
29. Which of them can act as oxidizing agent in acidic as well as in alkaline medium?
(A) $KMnO_4$ (B) $K_2Cr_2O_7$
(C) H_2O_2 (D) Cl_2

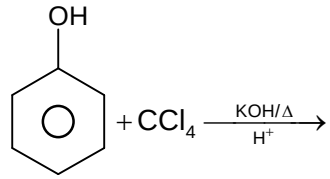
Space for Rough work

30. In which of the following reactions benzaldehyde will form in final product.

(A) 

(B) 

(C) 

(D) 

31. A dilute solution contains 't' moles of solute X in 1 kg of solvent with molal elevation constant K_b. The solute dimerises in the solution according to the following equation. The degree of association is α.

$$2X \rightleftharpoons X_2$$

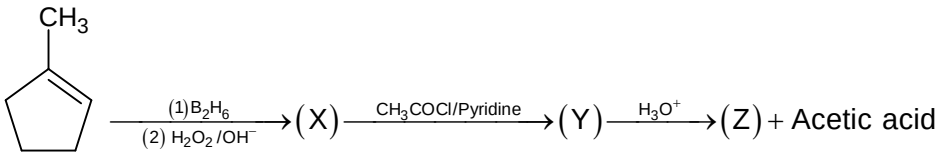
(A) The degree of association $\alpha = \dfrac{2(K_b t - \Delta T_b)}{K_b T}$

(B) The molecular mass observed will be lesser than actual molecular mass

(C) The colligative properties observed will be $\Delta T_{b\ obs} < \Delta T_{b\ theoretical}$; $\Delta T_{f\ obs} < \Delta T_{f\ theoretical}$; $\Delta P_{obs} < \Delta P_{theoretical}$

(D) The equilibrium constant for the process can be expressed as: $K = \dfrac{K_b(K_b t - \Delta T_b)}{[2\Delta T_b - K_b t]^2}$

32. The correct statements about the following reaction are



(A) (X) is transisomer of 2-Methylcyclopentan-1-ol

(B) (Y) is Cis isomer of ester

(C) (Z) is geometrical isomers of (X)

(D) (X) and (Z) have identical boiling point

Space for Rough work

Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 33 and 34

MnO_2 is the most important oxide of manganese. MnO_2 occurs naturally as the black coloured mineral pyrolusite. It is an oxidizing agent and decomposes to Mn_3O_4 on heating to 530^oC. It is used in the preparation of potassium permanganate and in the production of Cl_2 gas. Over half a million tons per year of MnO_2 is used in dry batteries.

33. In the laboratory, MnO_2 is made by
(A) Heating Mn in O_2
(B) Oxidizing Mn^{2+} in air
(C) Electrolytic oxidation of $MnSO_4$
(D) Precipitating MnO_2 from solution when performing titration of $KMnO_4$ in alkaline medium

34. In which of the following species, the colour is due to charge transfer.
(I) $[Mn(OH)_4]^{2-}$ (ii) MnO_4^{2-} (iii) MnO_2 (iv) $KMnO_4$
(A) I, II, III correct (B) II, IV are correct
(C) I, III are correct (D) only IV is correct

Space for Rough work

Paragraph for Question Nos. 35 and 36

Acid-base indicators are either weak organic acids or weak organic bases. Indicators change colour in dilute solution when the hydronium ion concentration reaches a particular value. For example, phenolphthalein is a colourless substance in any aqueous solution with pH less than 8.3. In between the pH range 8.3 to 10, transition of colour(colourless to pink) takes place and if pH of solution is greater than 10 solution is dark pink. Considering an acid indicator HIn, the equilibrium involving it and it's conjugate base(In^-) can be represented as:

$$\underset{\text{acidic form}}{HIn} \rightleftharpoons H^+ + \underset{\text{basic form}}{In^-}$$

pH of solution can be computed as: $pH = pK_{In} + \log\frac{[In^-]}{[HIn]}$

In general, transition of colour takes place in between the pH range, $pK_{In} \pm 1$.

35. An indicator is a weak acid and pH range is 4.0 to 6.0. If indicator in 50% ionized in a given solution, then what is the ionization constant of the acid?
(A) 10^{-4} (B) 10^{-5}
(C) 10^{-6} (D) None of these

36. Select the correct statement(s)
(A) at midway in the transition of an acidic indicator pH = pK_{In}
(B) Methyl orange(3.1 to 4.4) is suitable indicator for titration of weak acid and strong base
(C) Bromothymol blue(6.0 to 7.6) is not a good indicator for titration of HCl and NaOH
(D) Thymol blue(1.2 to 2.8) is a very good indicator for titration of 100 mL of 0.1 M NH_4OH (pK_b = 1.74) and 0.1 M HCl

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. If roots of quadratic equation $x^2 + (a^2 + 7)x - 15a = 0$ lie on either side of 2, then complete set of values of a is:

(A) (–2, 4) (B) $\left(-2, \frac{9}{2}\right)$

(C) $\left(-\frac{9}{2}, 2\right)$ (D) $\left(\frac{3}{2}, 6\right)$

38. Consider a triangle ABC and let a, b and c denote the lengths of the sides opposite to vertices A, B and C respectively. If $a = 1, b = 3$ and $C = 60^\circ$, then $\sin^2 B$ is equal to:

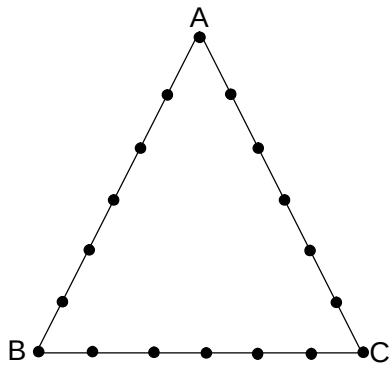
(A) $\frac{27}{28}$ (B) $\frac{3}{28}$

(C) $\frac{81}{28}$ (D) $\frac{1}{3}$

39. 18 points are indicated on the perimeter of a triangle ABC (see figure). How many triangles are there with vertices at these points?

(A) 331 (B) 408

(C) 710 (D) 711



Space for Rough work

40. If the mapping $f(x) = mx + c, m > 0$ maps [–1, 1] onto [0, 2] then $\tan\left(\tan^{-1}\frac{1}{7} + \cot^{-1}8 + \cot^{-1}18\right)$ is equal to:

(A) $f\left(\frac{2}{3}\right)$ (B) $f\left(\frac{1}{3}\right)$

(C) $f\left(\frac{-1}{3}\right)$ (D) $f\left(\frac{-2}{3}\right)$

41. $\lim\limits_{x\to1}\frac{nx^{n+1} - (n+1)x^n + 1}{(e^x - e)\sin\pi x}$ where $n = 100$ is equal to:

(A) $\frac{5050}{\pi e}$ (B) $\frac{100}{\pi e}$

(C) $-\frac{5050}{\pi e}$ (D) $-\frac{4950}{\pi e}$

42. Let $f(x) = \begin{cases} x^{3/5} & \text{if } x \le 1 \\ -(x-2)^3 & \text{if } x > 1 \end{cases}$ then the number of critical points on the graph of the function is:

(A) 1 (B) 2

(C) 3 (D) 4

43. The equation to the orthogonal trajectories of the system of parabolas $y = ax^2$ is:

(A) $\frac{x^2}{2} + y^2 = c$ (B) $x^2 + \frac{y^2}{2} = c$

(C) $\frac{x^2}{2} - y^2 = c$ (D) $x^2 - \frac{y^2}{2} = c$

Space for Rough work

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

44. If $2\cos\theta + \sin\theta = 1$, then the value of $4\cos\theta + 3\sin\theta$ is equal to:

(A) 3 (B) –5

(C) $\frac{7}{5}$ (D) –4

45. $\int \frac{x\,dx}{x^4 + x^2 + 1}$ equals:

(A) $\frac{2}{3}\tan^{-1}\left(\frac{2x^2+1}{\sqrt{3}}\right) + C$ (B) $\frac{1}{\sqrt{3}}\left[\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right) - \tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)\right] + C$

(C) $\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x^2+1}{\sqrt{3}}\right) + C$ (D) $\frac{1}{\sqrt{3}}\left[\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right) + \tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)\right] + C$

Where C is an arbitrary constant.

46. The co – ordinates of the point (s) on the graph of the function, $f(x) = \frac{x^3}{3} - \frac{5x^2}{2} + 7x - 4$ where the tangent drawn cut off intercepts from the co – ordinate axes which are equal in magnitude but opposite in sign, is:

(A) $\left(2, \frac{8}{3}\right)$ (B) $\left(3, \frac{7}{2}\right)$

(C) $\left(1, \frac{5}{6}\right)$ (D) none of these

47. Let $f(x) = \frac{x-1}{x^2}$ then which of the following is correct?

(A) $f(x)$ has minima but no maxima.

(B) $f(x)$ increases in the interval (0, 2) and decreases in the interval $(-\infty, 0) \cup (2, \infty)$.

(C) $f(x)$ is concave down in $(-\infty, 0) \cup (0, 3)$

(D) x = 3 is point of inflection

Space for Rough work

48. If $2xy\,dy = \left(x^2 + y^2 + 1\right)dx, y(1) = 0$ and $y(x_0) = \sqrt{3}$, then x_0 can be:

(A) 2 (B) –2
(C) 3 (D) –3

49. A line L passing through the point P (1, 4, 3), is perpendicular to both the lines $\frac{x-1}{2} = \frac{y+3}{1} = -\frac{z-2}{4}$ and $\frac{x+2}{3} = \frac{y-4}{2} = \frac{z+1}{-2}$.

If the position vector of point Q on L is (a_1, a_2, a_3) such that $(PQ)^2 = 357$, then $(a + a_2 + a_3)$ can be:

(A) 16 (B) 15
(C) 2 (D) 1

50. Solutions of the differential equation $\left(1 - x^2\right)\frac{dy}{dx} + xy = ax$ where $a \in R$, is:

(A) a conic which is an ellipse or a hyperbola with principal axes parallel to coordinates axes.
(B) centre of the conic is (0, a).
(C) length of one of the principal axes is 1
(D) length of one of the principal axes is equal to 2.

Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 51 and 52

Let $f(x) = x^2 + (a+2)x + a^2 - a + 2$. Given $a, \alpha, \beta (\alpha < \beta)$ be real numbers and α, β are the roots of the equation $f(x) = 0$.

51. If $\alpha < -2 < \beta$, then a lies in the interval:

(A) (0, 1) (B) (1, 2)
(C) (2, 3) (D) (3, 4)

Space for Rough work

52. If both roots of the equation $f(x) = 0$ lies in interval $(0, \infty)$, then range of a, is :

(A) $\left(\frac{2}{3}, 2\right)$ (B) $(-\infty, -2)$

(C) $(0, \infty)$ (D) ϕ

Paragraph for Question Nos. 53 and 54

Let f be an even function satisfying $f(x-2) = f\left(x + \left[\frac{6x^2+13}{x^2+2}\right]\right) \forall x \in R$ and

$$f(x) = \begin{cases} 3x, & 0 \le x < 1 \\ 4 - x, & 1 \le x \le 4 \end{cases}$$

[**Note:** [y] denotes greatest integer function of y.]

53. The area bounded by the graph of $f(x)$ and the x – axis from x = –1 to x = 9 is:

(A) $\frac{31}{2}$ (B) 15

(C) 12 (D) $\frac{15}{2}$

54. The value of $f(-89) - f(-67) + f(46)$ is equal to :

(A) 4 (B) 5

(C) 6 (D) 7

Space for Rough work